

# Resource allocation in 5G cloud-RAN using deep reinforcement learning algorithms: A review

Mohsen Khani<sup>1</sup> | Shahram Jamali<sup>2</sup> | Mohammad Karim Sohrabi<sup>1</sup> |  
 Mohammad Mohsen Sadr<sup>3</sup> | Ali Ghaffari<sup>4,5</sup>

<sup>1</sup>Department of Computer Engineering,  
 Semnan Branch, Islamic Azad University,  
 Semnan, Iran

<sup>2</sup>Department of Computer Engineering,  
 University of Mohaghegh Ardabili,  
 Ardabil, Iran

<sup>3</sup>Department of Computer and  
 Information Technology Engineering,  
 Payame Noor University, Tehran, Iran

<sup>4</sup>Department of Computer Engineering,  
 Tabriz Branch, Islamic Azad University,  
 Tabriz, Iran

<sup>5</sup>Department of Computer Engineering,  
 Faculty of Engineering and Natural  
 Science, Istinye University, Istanbul,  
 Turkey

## Correspondence

Mohsen Khani, Department of Computer  
 Engineering, Semnan Branch, Islamic  
 Azad University, Semnan, Iran.  
 Email: [mohsenkhani622@gmail.com](mailto:mohsenkhani622@gmail.com)

## Abstract

This paper reviews recent research on resource allocation in 5G cloud-based radio access networks (C-RAN) using deep reinforcement learning (DRL) algorithms. It explores the potential of DRL for learning complex decision-making policies without human intervention. The paper first introduces the C-RAN architecture and resource allocation concepts, followed by an overview of DRL algorithms applied to C-RAN. It discusses the challenges and potential solutions in applying DRL to C-RAN resource allocation, including scalability, convergence, and fairness. The review concludes by highlighting open research directions for future investigation. By providing insights into the state-of-the-art techniques for resource allocation in 5G C-RAN using DRL, this paper emphasizes their potential impact on advancing 5G network technology.

## 1 | INTRODUCTION

The advent of 5G technology has brought about a transformative era, revolutionizing communication and connectivity with its limitless possibilities.<sup>1</sup> With its lightning-fast data transfer speeds and near-instantaneous connections, 5G promises to enable innovative emerging technologies such as self-driving vehicles, remote medical procedures, and immersive virtual reality experiences.<sup>2–4</sup> However, the deployment of 5G networks requires a new approach to network architecture and resource allocation due to the significant increase in traffic volume and complexity.<sup>5</sup>

One promising solution to this demand is the cloud radio access network (C-RAN), which decouples the RAN functions from traditional base stations and centralizes them in a cloud-based processing unit. Figure 1 illustrates the C-RAN architecture, which offers a versatile and efficient resource allocation mechanism that caters to the varied demands of 5G applications. C-RAN has been recently implemented to support various emerging services, such as the Internet of Things (IoT), Industrial IIoT, Vehicle-to-Everything (V2E), Vehicle-to-Vehicle (V2V) communications,<sup>6</sup> and Mobile Edge Computing (MEC).<sup>8–10</sup> These services<sup>7–9</sup> require high-speed connectivity, low latency, and reliable communication, which can be achieved through C-RAN technology.

In C-RAN, base stations are centralized in cloud infrastructure, providing computational and storage resources for processing the large amounts of data generated by these services. This enables efficient network management and faster

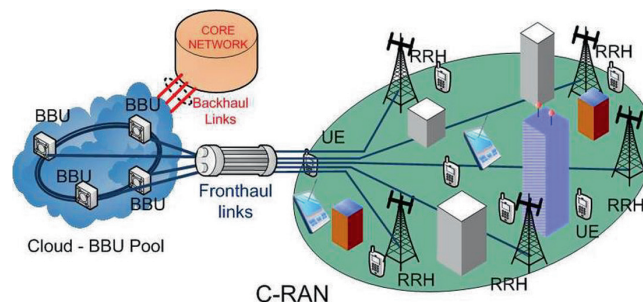


FIGURE 1 Cloud-RAN-architecture.<sup>13</sup>

response times for critical applications. Additionally, C-RAN supports network slicing, allowing for the creation of virtual networks tailored to specific services and their requirements.

Resource allocation (RA) in C-RAN involves strategically assigning and optimizing resources to tailor the user experience according to their distinct quality-of-service (QoS) requirements. This intricate orchestration aims to accommodate the diverse spectrum of individuals while maximizing overall system capacity and energy efficiency.<sup>10,11</sup> Traditional RA schemes rely on mathematical optimization techniques that require accurate channel state information and predefined system models. However, these strategies may lack the scalability needed to address the ever-changing and heterogeneous attributes of 5G. An alternative approach to RA in C-RAN is leveraging deep reinforcement learning (DRL) algorithms, which utilize machine learning techniques to learn complex decision-making policies directly from raw data without explicit human intervention.<sup>12</sup>

DRL algorithms have shown promising results in solving RA problems in wireless communication networks, including C-RAN. The proposed approach in this paper mainly focuses on a comprehensive survey analysis of the current state-of-the-art DRL-based RA approaches in C-RAN. We introduce the fundamental concepts of C-RAN architecture and RA, as well as discuss the potential solutions and challenges in applying DRL to C-RAN RA. By doing so, we aim to provide insights into cutting-edge methodologies for RA in 5G C-RAN using DRL algorithms and their potential impact on the development of 5G networks.

For better understanding, Table 1 provides a comprehensive list of essential abbreviations and definitions relevant to the subject matter.

The subsequent part of the article is organized as follows: Section 2 presents a summary of contemporary investigations into resource distribution within 5G C-RAN, including an introduction to DRL algorithms and their potential for addressing intricate decision-making challenges. This section also discusses cutting-edge DRL algorithms applied to RA. In Section 3, we address the challenges and solutions related to applying DRL algorithms to C-RAN RA, including scalability issues, convergence challenges, and ensuring fairness in RA. Section 4 identifies open areas for future investigation in the application of DRL to C-RAN RA and explores the possible influence of DRL on the advancement of 5G networks. Finally, Section 5 offers a conclusion.

## 2 | LITERATURE REVIEW

### 2.1 | Overview of traditional methods on RA in 5G C-RAN

In the last decade, there has been a significant increase in demand for high-speed and low-latency communication services. As a result, the emergence of next-gen wireless networks has come to fruition, including 5G. One promising architecture for 5G networks is the C-RAN, which utilization of centralized baseband processing and efficient RA stands as a critical aspect within C-RAN, directly influencing network performance and enhancing energy efficiency. Various methods have been suggested to enhance the efficiency of C-RANs in terms of RA, including power distribution strategies,<sup>14,15</sup> user association,<sup>16–18</sup> and virtual machine placement.<sup>19,20</sup> These advanced methodologies strive to enhance network capacity, mitigate latency, and optimize energy efficiency by accounting for the dynamic and heterogeneous nature of the network's inherent traits.<sup>21</sup> Traditional research on RA in 5G C-RAN has focused on several key areas, including:

**TABLE 1** A list of important abbreviations related to the study.

| Acronym | Abbreviation                                            | Acronym | Abbreviation                             |
|---------|---------------------------------------------------------|---------|------------------------------------------|
| 5G      | The fifth generation of wireless communication networks | EH      | Energy harvesting                        |
| DRL     | Deep reinforcement learning                             | A3C     | Asynchronous Advantage Actor-Critic      |
| MEC     | Mobile Edge Computing                                   | RANs    | Radio Access Networks                    |
| DTF     | Delay/time to finality                                  | QoE     | Quality of experience                    |
| VBS     | Virtual Base Station pool                               | SDMN    | Software-defined mobile networks         |
| RRH     | Remote Radio Heads                                      | BS      | Base stations                            |
| GHN     | Front-haul network                                      | CoMP    | Coordinated Multipoint                   |
| IOCS    | Inter-operator cooperation scheme                       | URLLC   | Ultra-reliable low-latency communication |
| DQN     | Deep Q-Networks                                         | AC      | Admission Control                        |
| H-CRAN  | Heterogeneous Cloud Radio Access Networks               | F-Aps   | Fog access points                        |
| OFDM    | Orthogonal frequency division multiplexing              | MINP    | Mixed-integer non-linear programming     |
| SE      | Spectral efficiency                                     | IoRT    | Robotic Things                           |
| SLA     | Service level agreement                                 | WMMSE   | Weighted MMSE                            |
| ES      | Edge server                                             | MCA     | MsgA channel allocation                  |
| MSE     | Mean squared error                                      | NN      | Neural network                           |
| BAC     | Backscatter communication                               | LEO     | Low earth orbit                          |
| MARL    | Multi-agent reinforcement learning                      | NOMA    | Non-orthogonal multiple access           |
| MTC     | Machine Type Communication                              | DC      | Dual connectivity                        |
| E2E     | End-to-end                                              | LEO     | Low earth orbit                          |
| VNF     | Virtual network function                                | NOMA    | Non-orthogonal multiple access           |
| DC      | Dual connectivity                                       |         |                                          |

### 2.1.1 | Simultaneous fine-tuning of radio and computational resources to achieve optimal performance

The central objective of this approach is centered on optimizing the allocation and distribution of both radio frequency (RF) and computational resources, with the primary aim of mitigating latency and maximizing the overall efficiency of the network. To achieve this goal, researchers have developed a variety of algorithms. The paper<sup>22</sup> introduces a notable algorithm that addresses the challenge of mitigating the computational overhead on mobile devices through the offloading of resource-intensive tasks to cloud servers, resulting in a significant reduction in the burden placed upon these devices. This strategy enhances the computing capabilities of mobile devices while simultaneously preserving battery life. The investigation centers around a MIMO multicell framework wherein a multitude of mobile users actively pursue computation offloading services from a centrally located cloud server. The authors address the challenge of offloading by formulating a combined optimization framework for managing radio and computational resources. The main goal is to reduce overall energy usage while maintaining adherence to latency limitation. Considering the non-linear characteristics of the optimization problem, the authors put forth an iterative algorithm employing a novel approach known as the successive convex approximation technique. By adopting this algorithmic approach, it becomes possible to implement a distributed and parallel framework that spans numerous radio access points. This, in turn, minimizes the requirement for extensive coordination or signaling from the cloud, enhancing system efficiency and reducing latency.

In Reference [23], the authors proposed a joint optimization framework for Mobile Edge Computing (MEC)<sup>24</sup> that utilizes blockchain technology. Existing works on blockchain and MEC are often designed separately, resulting in sub-optimal performance. The proposed framework encompasses the integration of MEC and blockchain services, to attain an optimal equilibrium in terms of energy consumption and time required to achieve finality. A comprehensive evaluation is conducted to directly compare the performance of the MEC system and the blockchain system, specifically in

terms of their time-to-finality metrics. The authors devise an optimization problem that simultaneously optimizes user association, data rate allocation, block producer scheduling, and computational RA. To efficiently design algorithms, they decouple the optimization variables. The authors develop iterative and bisection algorithms for user association and data rate allocation, as well as computing RA, respectively.

### 2.1.2 | Virtualization-based resource management

In 5G C-RAN, virtualization technology is widely used to efficiently allocate and share resources among different virtual networks. Various resource management schemes based on virtualization have been proposed to improve network performance, such as dynamic resource pooling and adaptive RA. In Reference [25], the C-RAN structure is divided into three components: a Virtual Base Station (VBS) pool, Remote Radio Heads (RRH), and a Front-haul network. The VBS pool comprises multiple baseband units that employ real-time cloud computing technology for digital processing tasks. This type of architecture facilitates the exchange of control data among various operators and promotes collaboration to deliver top-notch 5G services. Nevertheless, establishing a secure 5G ecosystem requires reliable collaboration and platform authentication based on trust among multiple mobile operators. To achieve trustworthy and secure resource sharing among these operators, the scholars propose a C-RAN inter-operator cooperation scheme (IOCS) that relies on trusted computing platforms. Furthermore, they create a reliable collaborative framework utilizing OpenStack to deploy IOCS and carry out an assessment of performance and simulation analysis to showcase its enhanced operational effectiveness in comparison to alternative approaches.

By the proliferating utilization of mobile devices and the exponential upsurge in mobile internet traffic, there exists a growing need for uninterrupted and high-capacity cellular access, as extensively explored in an additional scholarly article referenced as [26]. To tackle this issue, researchers have employed two main approaches: centralized cloud processing through virtualization and the segregation of the shift in ownership of infrastructure from service provisioning within cellular systems. To facilitate inter-operator resource sharing within C-RAN infrastructure, the authors introduce a novel auction-based method. In this approach, each operator submits bids for a specific sub-network of the C-RAN infrastructure. Taking inspiration from auctions employing Maximal-in-Range methodologies and advanced graph methods, the authors have developed and evaluated a computationally feasible C-RAN resource auction that ensures truthfulness and achieves nearly optimal social welfare.

### 2.1.3 | QoS-aware RA

QoS is critical in 5G C-RAN, especially for delay-sensitive applications. Recent research has focused on developing QoS-aware RA algorithms that can guarantee the required QoS metrics while maximizing network throughput and minimizing resource utilization. In Reference [27], the authors proposed a method to optimize RA in C-RAN using cooperative game theory. Through the employment of a weight-based policy that takes into consideration real-time demands and the relative importance of executed services, baseband units are prioritized to ensure fairness. This is achieved by distributing resources among the baseband units based on their respective weights within the resource pool, thereby fostering equitable allocation and optimizing resource utilization. In Reference [28], the repercussions of inefficient network resource utilization within a dynamic traffic environment are investigated, encompassing issues such as load imbalances, heightened call-blocking occurrences and compromised QoS. The authors investigate load balancing solutions using the dynamic re-mapping ability of C-RAN to configure RRHs to proper base band unit sectors. They introduce the SOCRAN algorithm as a solution to enhance network QoS by effectively balancing traffic load and minimizing handovers within the network. The optimization problem is tackled using a combination of genetic algorithm (GA) and discrete particle swarm optimization (DPSO).

### 2.1.4 | Energy efficiency (EE)

EE is another important aspect of 5G C-RAN RA. Various energy-efficient RA schemes have been proposed to reduce power consumption while maintaining network performance, such as sleep mode optimization, base station clustering,

**TABLE 2** Brief traditional techniques for efficient RA in C-RAN.

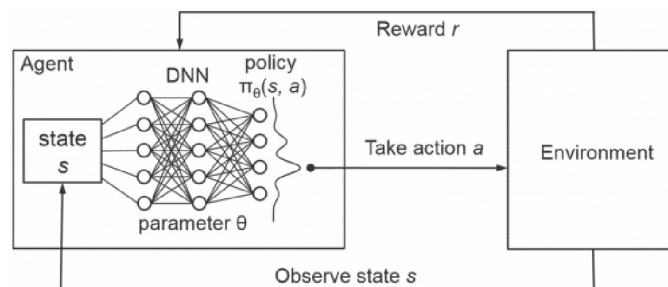
| Resource allocation technique                           | Key objectives                                                                 | Key features/components                                                                                                                                             |
|---------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Power allocation (PA)                                   | Improve network capacity, reduce energy consumption                            | Allocate power resources to different base station units                                                                                                            |
| Joint optimization of radio and computational resources | Minimize latency, maximize network efficiency                                  | Enabling task offloading of computationally intensive operations from mobile devices to cloud servers for integrated radio and computational resource optimization. |
| Virtualization-based resource management                | Efficient allocation and sharing of resources among different virtual networks | Dynamic resource pooling, adaptive RA                                                                                                                               |

and smart antenna selection. In Reference [29], the authors proposed an algorithm that optimizes BBU-RRH mapping and user association to save on system costs while ensuring QoS. The proposed algorithm performs well in simulations.

In Table 2, we made a brief classification of the traditional methods in the field of RA in C-RAN architecture.

## 2.2 | Introduction to DRL algorithms and their potential for solving complex decision-making problems

As shown in Figure 2, the structure of DRL is illustrated. DRL is a type of ML algorithm that has become increasingly popular in recent times owing to its capability to acquire knowledge of intricate tasks through interaction with the environment.<sup>30,31</sup> DRL combines DNN, which are powerful function approximators, and reinforcement learning, which is a paradigm for learning by engaging in iterative interactions with an environment and learning from experience.<sup>32–34</sup> DRL operates on the fundamental concept of training an agent to make decisions within an environment to maximize a cumulative reward over the long term. The agent receives observations from the environment and employs them to determine the most appropriate course of action.<sup>35,36</sup> The choice of action is based on a policy, which is represented by a deep neural network.<sup>37</sup> The policy takes the observations as input and outputs a distribution over possible actions. The training process involves running the agent in the environment and updating the policy parameters to maximize the expected future reward.<sup>38</sup> This is done using a variant of stochastic gradient descent called policy gradient, which computes gradients of the expected future reward concerning the policy parameters. The policy parameters are updated using gradients, which guide the optimization process toward increasing the expected future reward.<sup>39</sup> To estimate the anticipated future reward, a discount factor is used to give more weight to immediate rewards than to future rewards. This helps the agent to focus on short-term goals while still considering long-term goals.<sup>40</sup> The selection of the neural network structure relies on the characteristics of the environment and the specific task being performed.<sup>41–43</sup> The network typically consists of multiple layers, with each layer performing a different type of computation.<sup>44,45</sup> The hidden layers are designed to extract features from the observations that are relevant to the task.<sup>46</sup> The choice of loss function also depends on the task at hand. In numerous instances, the loss function corresponds to the negative logarithm of the agent's chosen

**FIGURE 2** Structure of DRL.<sup>55</sup>

actions, thereby multiplied by the discounted future reward.<sup>47</sup> This is known as the policy gradient loss.<sup>48</sup> One advantage of DRL in public environments, especially wireless networks, is its ability to adapt to changing conditions.<sup>49</sup> In wireless networks, the dynamic nature is a result of various factors including interference, mobility, and fluctuating traffic demands.<sup>50</sup> Through continuous interaction with the environment and policy updates, DRL techniques can effectively adapt to these changes.<sup>51</sup> DRL demonstrates an additional benefit through its capability to learn complex tasks that are difficult to program directly.<sup>52</sup> In wireless networks, many tasks are difficult to optimize through traditional methods, such as load balancing and spectrum management.<sup>53</sup> DRL can learn to optimize these tasks by trial and error, without requiring explicit programming.<sup>54</sup> In conclusion, DRL is a powerful machine learning algorithm that combines DNN and reinforcement learning to learn complex tasks by interacting with an environment. DRL has many advantages in public environments, especially wireless networks, where the environment is constantly changing, and traditional optimization methods may not be effective.

## 2.3 | Cutting-edge DRL algorithms implemented within 5G C-RAN scenarios

### 2.3.1 | Deep Q-network (DQN)

DQN is a class of RL algorithms that leverage NN for approximating the Q function, as demonstrated in research papers.<sup>32,56</sup> The Q function estimates the long-term reward for taking a particular action in a given state. DQNs have been useful in solving problems with large and complex action spaces.<sup>57–59</sup> In Reference [60], the authors proposed a DQN method to reduce power usage in Heterogeneous C-RAN (H-CRAN). The scientists have developed a systematic approach to enable autonomous activation of cellular functions and customized resource allocation techniques for achieving a harmonious balance between energy consumption, environmental preservation, and ensuring the satisfactory quality of service. In Reference [44], the authors discuss the utilization of the DQN and deep deterministic policy-gradient (DDPG) algorithms proposed to optimize the allocation of resources within intelligent reflecting surface (IRS) enhanced orthogonal frequency division multiplexing (OFDM) systems. The proposed approach involves the simultaneous optimization of subcarrier allocation, base station transmit beamforming, and phase shift adjustments of the IRS to maximize the overall system sum rate. In Reference [61], the authors explore the potential of DQN resource orchestration techniques in managing and allocating resources within a RAN that contains IIoT devices, such as base stations that share physical infrastructure. The authors introduce generative adversarial network-based deep distributional noisy Q-networks (GAN-NoisyNet) and dueling GAN-NoisyNet to achieve variable service demands and maximize the optimal policy for IIoT<sup>62</sup> reward by optimizing system throughput, spectral efficiency (SE), service level agreement (SLA), optimizing both power consumption and transmission delay while maintaining a high packet transmission rate. By conducting extensive simulations, the proposed algorithms demonstrate significant improvements in RA within IIoT networks.

In Reference [63], the authors proposed a method for intelligent resource management in C-RAN using DQL. The method aims to improve QoS under limited communication resources by maximizing efficient usage of spectrum and bandwidth resources. The proposed method provides a more subdivided allocation policy, resulting in higher performance and improved efficiency compared to other standard solutions. In summary, DQNs use NN to estimate long-term rewards for actions in a given state and are effective in solving problems with large and complex action spaces. DQNs can minimize energy consumption, optimize RA systems, and manage resources in C-RAN. Also, the computational complexity of the DQN algorithm depends on several factors, including the size of the neural network, the number of training iterations, and the complexity of the environment or problem being solved. In DQN, the most computationally expensive operations occur during the training phase, specifically during the forward and backward propagation steps. During forward propagation, the neural network evaluates the current state and predicts the Q-values for all possible actions. This involves passing the state through multiple layers of the network, performing matrix multiplications, and applying activation functions. The computational complexity of forward propagation is typically proportional to the number of neurons and layers in the network, denoted as  $O(n)$ , where  $n$  represents the number of parameters in the network. During backward propagation, the network updates its weights based on the error between the predicted Q-values and the target Q-values. This involves computing the gradients of the loss function concerning the network parameters and applying gradient descent optimization. The computational complexity of backward propagation is also proportional to the number of parameters in the network, denoted as  $O(n)$ . In addition to forward and backward propagation, DQN also involves other operations such as experience replay, which stores and samples previous experiences for training, and target network updates, which periodically copy the current network weights to a separate target network. Overall, the



6. Backpropagate the loss through the Q-network and update the network weights to minimize the loss.
7. Repeat steps 3–4: Continue this process of selecting actions, updating the Q-network, and collecting experience for a certain number of iterations or until a stopping criterion is met. By iteratively updating the Q-network using this process, the algorithm learns to approximate the optimal action-value function (Q-function) for the given environment.

### 2.3.2 | Policy gradient

Policy gradient techniques optimize the policy function, which establishes a relationship between states and corresponding actions. This is in contrast to value-based methods like DQN, which provide an assessment of the worth in a specific state-action combination.<sup>65</sup> Policy gradient approaches tend to exhibit enhanced efficacy when dealing with scenarios involving continuous action spaces.<sup>66</sup> The policy gradient algorithm has been used in several articles for RA in 5G networks.<sup>63,67,68</sup> The policy gradient algorithm for RA in 5G networks offers a significant benefit by effectively managing intricate and ever-changing surroundings, all while considering rewards over extended periods. The algorithm can learn from past experiences and adapt its actions accordingly to optimize the anticipated long-term benefit. Additionally, the use of DRL techniques, such as NN, can help capture the complex relationships between various network parameters and optimize the RA process. In summary, the policy gradient algorithm offers a promising approach for RA in 5G networks, providing the ability to handle complex environments and learn from past experiences to make optimal decisions. The computational complexity of the Policy Gradient algorithm is  $O(N * T * d)$ , here  $N$  is the number of iterations,  $T$  is the time steps in each iteration, and  $d$  is the dimensionality of the policy's parameter space. More concisely, the computational complexity of the Policy Gradient algorithm is  $O(N * T * d)$ .

### 2.3.3 | Actor-critic

Figure 4 shows how binary method works. This method combines aspects of approaches based on value and policy to RL. The “actor” component of the algorithm learns to choose actions based on the current policy, while the “critic” component evaluates the quality of the policy.<sup>69,70</sup> Actor-critic methods have proven to be effective in solving problems that involve continuous action spaces.<sup>39</sup> For example, the Actor-Critic algorithm has been applied in various scenarios to solve RA and optimization problems.<sup>71</sup>

In Reference [72], the researchers suggest a computational framework grounded in DRL to tackle the issue of devising efficient strategies for offloading computations and RA policies within an MEC system. The primary goal is to reduce the overall expenses associated with delays and energy usage, which is achieved by estimating the value function of the critics using DNNs and representing a parametrized stochastic strategy in the actor part. The simulation outcomes indicate that this approach considerably diminishes the overall expenses in comparison to alternative approaches.

In Reference [69], the authors primarily concentrate on enhancing user scheduling and resource allocation in HetNets powered by hybrid energy sources, with the ultimate aim of optimizing energy consumption and maximizing overall network efficiency. To achieve this, they employ a model-free RL approach, which entails acquiring the most effective strategy by engaging with the surroundings. A parameterized policy, utilizing a Gaussian distribution, is introduced as the

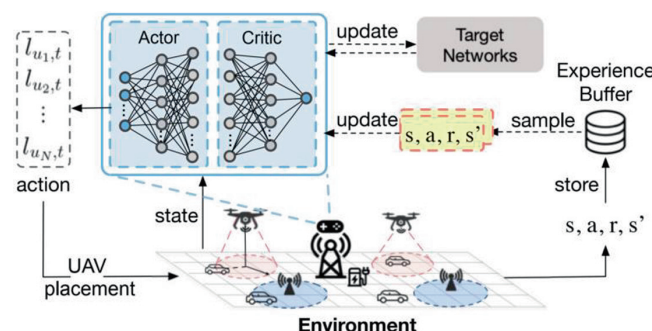


FIGURE 4 The performance of the Actor-critic method.<sup>64</sup>

policy-gradient-based actor-critic algorithm to produce stochastic actions continuously. The critic module evaluates the policy's effectiveness and assists the actor in acquiring the gradient of the policy by leveraging the advantage function. The proposed algorithm is evaluated through numerical simulations, demonstrating its convergence properties and analyzing the network's energy efficiency.

In Reference [73], the authors tackle the challenges of content caching strategy, computation offloading policy, and radio RA in fog-enabled IoT systems. Their approach focuses on jointly optimizing these aspects using an RL-based actor-critic learning framework. The primary objective is to minimize the average end-to-end delay in a stochastic environment. To estimate the value functions in the critic component, DNNs are utilized as function approximators. Additionally, another DNN is employed in the actor part to represent a parameterized stochastic policy, which is continually improved with assistance from the critic. The researchers leverage the Natural policy gradient method to prevent convergence to local maxima. Numerical simulations are conducted to demonstrate the algorithm's learning capability and analyze the resulting end-to-end service latency.

Overall, the Actor-Critic algorithm has proven to demonstrate efficacy in addressing RA and optimization problems in various scenarios, including multi-user MEC systems, HetNets with hybrid energy supply, and fog-enabled IoT networks.<sup>74</sup> The use of DNN and model-free approaches has enabled efficient and intelligent management of network resources and improved network performance. The computational complexity of the Actor-Critic algorithm is generally summarized as  $O(k * n)$ , where  $k$  represents the number of iterations and  $n$  represents the number of parameters in the model. Please note that this is a simplified representation of the computational complexity, as the actual complexity can vary depending on the specific implementation and variations of the algorithm.

### 2.3.4 | Asynchronous advantage Actor-critic (A3C)

A3C is a type of actor-critic algorithm that has been distributed. The "asynchronous" component indicates that several agents are learning independently and in parallel asynchronously. The term "advantage" refers to the use of the advantage function, which measures the comparative effectiveness of an action against the typical action.<sup>75</sup> In various network architectures, A3C has been used for different RA problems. In one paper,<sup>76</sup> A3C was implemented for optimizing RA in 5G for RAN slicing. The authors suggested an energy-efficient RA approach assisted by deep reinforcement learning (EE-DRL-RA) to simultaneously assign power and resource units while maintaining slice separation with minimal computational and temporal intricacy. They employed DL for making decisions regarding RA over an extensive time frame and employed DRL for making decisions regarding RA over a shorter time frame. A3C served as the DRL method along with stacked and bidirectional long-short-term-memory (SBI LSTM) networks, which were used as supervised deep learning methods. Based on the simulation outcomes, EE-DRL-RA demonstrated superior performance compared to a state-of-the-art approach in terms of convergence rate, computational intricacy, energy efficiency, user acceptance rate, and inter-slice isolation level. In Reference [77], the authors employed A3C to enhance energy efficiency and ensure optimal quality of experience (QoE) measurements in video streaming over software-defined mobile networks (SDMN) integrated with MEC. The authors proposed a novel mechanism to jointly consider buffer dynamics, video quality adaptation, edge caching, video transcoding, and transmission. They used A3C to solve the corresponding MDP issues. The simulation results showed that the proposed scheme could achieve energy-saving and QoE enhancement goals while satisfying the corresponding constraints. Overall, implementing A3C in these RA problems has exhibited promising outcomes, including optimizing resource utilization, enhancing network performance, reducing energy consumption, and improving user experience. The time complexity of the A3C algorithm depends on the number of training iterations and the size of the neural network used for the actor and critic models. It involves performing forward and backward passes through the network for each iteration, which can be computationally expensive. However, the A3C algorithm benefits from its asynchronous nature, where multiple threads or processes can update the network parameters simultaneously. This parallelization helps distribute the computational load and potentially speed up the training process. Announcement: The exact computational complexity of the A3C algorithm can vary depending on implementation details and hardware resources.

### 2.3.5 | Proximal policy optimization (PPO)

PPO is a type of policy gradient algorithm that optimizes the objective function subject to a constraint on the amount by which the policy can change. This ensures that the policy update does not deviate too far from the previous policy.<sup>78</sup>

In the paper,<sup>79</sup> the authors propose a framework to allocate resources for base stations and mobile devices, using a multi-objective PPO algorithm to solve a multi-objective programming problem. This approach surpasses conventional methods in discovering superior solutions for resource allocation across various service types. In Reference [80], the authors address the challenge of providing 5G services in an automated warehouse environment by utilizing swarm robotics<sup>81–83</sup> controlled using an industrial controller over the 5G network. They suggest a game-theoretic Coordinated Multipoint clustering algorithm integrated with PPO to manage the warehouse robots responsible for storing goods. In Reference [84], the authors proposed a learning-based resilience guarantee framework for multi-UAV collaborative QoS management during epidemics or disasters. They use PPO to solve a partial observable MDP problem and show that policies trained with PPO have better performance compared to other algorithms when UAVs have local communication capabilities. The computational complexity of the PPO algorithm is typically considered to be linear,  $O(n)$ , concerning the number of training iterations. However, it is not possible to provide a numerical calculation of the exact computational complexity without specific details and measurements of the implementation, environment, and network size.

### 2.3.6 | Deep deterministic policy gradient (DDPG)

DDPG is an actor-critic method designed for solving continuous control problems. Specifically, the “actor” outputs a deterministic action given the current state, while the “critic” evaluates the quality of that action. In recent research, DDPG has been employed as a key algorithm in developing solutions for RA and computation offloading in various communication networks. In Reference [85], a federated DRL-based algorithm is proposed to address the dynamic computation offloading and RA problem in Fog Access Points (F-APs). The DDPG algorithm is used by each F-AP to solve this problem effectively. Federated learning is utilized to train the DDPG agents, reducing computing complexity and protecting user privacy. Simulation results show that the proposed algorithm can achieve lower task execution delays and energy consumption for mobile devices compared to other strategies. Similarly, in Reference [86], a DDPG-based framework is proposed to jointly optimize computation task offloading and RA in F-RANs. The authors address the challenge of enhancing system processing efficiency in the F-RAN through a comprehensive optimization approach encompassing offloading mode selection, channel allocation, power control, and computation RA. The proposed DDPG algorithm significantly reduces the computing complexity and signaling overhead of the training process, as compared to centralized learning-based approaches. The simulation results provide compelling evidence of substantial advancements in system processing efficiency and reduction in task latency, surpassing the performance of other benchmarks. Moreover, in Reference [87], to efficiently address the mixed-integer non-linear programming (MINP) problem and learn system dynamics, a novel hierarchical multi-agent DRL framework named DeepRAT is introduced. The authors focus on tackling the challenge of maximizing the downlink sum rate while considering cost factors in next-generation HetNets through the optimal joint assignment of Radio Access Technologies (RATs) and PA. The framework comprises two primary stages: the RATs-EDs assignment stage, employing a Deep Q-Network (DQN) algorithm as a single-agent approach, and the PA stage, utilizing a multi-agent Deep Deterministic Policy Gradient (DDPG) algorithm. The simulation results demonstrate the superiority of the proposed algorithm over existing heuristic methods in terms of network utility. Additionally, it exhibits rapid and adaptive responses to sudden alterations in network dynamics. In conclusion, recent studies have shown that DRL algorithms, particularly DDPG, are effective in optimizing task execution delay, energy consumption, and system processing efficiency in communication networks. These studies demonstrate the effectiveness of using DDPG in DRL for end-to-end network slicing RA, joint computation offloading and RA, RATs assignment, and PA. The computational complexity of the Deep Deterministic Policy Gradient (DDPG) algorithm is typically considered to be quadratic,  $O(n^2)$ , concerning the size of the neural network being used. This means that as the network size increases, the computational requirements of DDPG will increase exponentially. However, it is worth noting that the exact computational complexity can vary depending on factors such as the specific implementation details and the environment being trained on.

### 2.3.7 | Twin delayed deep deterministic policy gradient (TD3)

TD3 is an algorithm that builds upon the success of its predecessor, DDPG. TD3 addresses some of the limitations of DDPG by reducing overestimation and variance in the  $Q$ -value estimates. One key feature of TD3 is the use of two critics instead of one, which helps to better estimate the  $Q$ -values. Additionally, TD3 applies random noise to the target policy

during training, which encourages exploration and can lead to better generalization.<sup>88</sup> In recent years, TD3 has been applied to a variety of real-world problems with promising results. For example, in reference [89], the authors propose a TD3-based controller for the PA in the Internet of Robotic Things (IoRT) network. The proposed algorithm outperforms model-based methods like fractional programming (FP) and weighted MMSE (WMMSE), as well as model-free algorithms like DQN and DDPG, in terms of sum-rate performance and generalization power. Another application of TD3 is in improving the access success probability of 5G-and-beyond low earth orbit (LEO) satellite communication systems. In Reference [90], The writers suggest a controller based on TD3 for allocating power in networks of Internet of Robotic Things (IoRT). The algorithm they propose performs better than model-based methods such as fractional programming (FP) and weighted MMSE (WMMSE), as well as model-free algorithms like DQN and DDPG, in terms of total data transmission rate and generalization ability. TD3 has also been used to enhance the likelihood of successful access in low earth orbit (LEO) satellite communication systems that employ 5G-and-beyond technology. In Reference [90], the writers propose an adaptive MsgA channel allocation (MCA) strategy which is implemented via an enhanced TD3-based DRL model. This approach adjusts the mapping relationship of channels on MsgA through historical access information, leading to a significant increase in the probability of successful access. In addition to these applications, TD3 has been employed in creating innovative computation offloading schemes in MEC systems, as described in Reference [91]. Here, the writers focus on optimizing task segmentation and energy allocation together to minimize total energy consumption. They form the joint optimization problem into a MINLP problem and suggest a new heuristic algorithm to achieve a suboptimal solution. Then, they propose an online learning algorithm based on TD3 to fulfill user mobility demands. The suggested method outperforms other techniques, and the TD3-based algorithm reduces the execution time compared to the heuristic algorithm. In Reference [92], the researchers introduce a well-balanced and efficient approach for RA in a wide range of MTC (Massive Machine Type Communications) services. They employ a multi-agent DRL (MDRL) technique to address the scheduling of E2E virtual network functions (VNF), as well as the allocation of resources across core network nodes, links, and access network subcarriers, employing various strategies. To facilitate the application of the DRL algorithm, they propose the ASgn mapping scheme, which simplifies compound variables into continuous variables within the optimization problem. Subsequently, they put forward the Model Paralleling Multiagent Twin Delayed Deep Deterministic (MPMA-TD3) policy gradient algorithm, which effectively handles the challenges posed by massive services while simultaneously reducing training time and action space for agents. Moreover, they enhance the MPMA-TD3 algorithm to cater to diverse services, resolve issues related to unequal RA, and achieve a well-balanced distribution of resources across all services. Through simulations, the results demonstrate that the proposed algorithms surpass alternative approaches in terms of reward, delay, cost, and training time for massive services. Notably, the Improved MPMA-TD3 algorithm exhibits exceptional capabilities in achieving service equilibrium. The computational complexity of TD3 algorithm is generally described as quadratic,  $O(n^2)$ , about the size of the neural network employed. Consequently, as the size of the network grows, the computational demands of TD3 experience exponential growth. Nevertheless, it is crucial to acknowledge that the precise computational complexity can differ based on multiple factors, including specific implementation details and the training environment.

### 2.3.8 | Soft Actor-critic (SAC)

SAC is an actor-critic method that seeks to maximize the expected reward while maintaining a minimum level of entropy in the policy. The term “soft” refers to the fact that SAC optimizes a lower bound on the standard RL objective, which includes an entropy term to encourage exploration. In recent years, SAC has been applied to various RA problems in wireless networks.<sup>93,94</sup> For instance, in Reference [95], the authors propose a MEC system for cooperative computation offloading among edge-cloud and edge-edge servers.<sup>96</sup> To support services with minimal delays, the authors frame the problem as an MDP and put forth two intelligent computation offloading algorithms based on SAC, namely centralized SAC offloading and decentralized SAC offloading. The results of their evaluation demonstrate that their proposed algorithms outperform existing methods in terms of service latency. Similarly, in Reference [97], the authors introduce an effective model-free approach based on SAC for dynamic resource scheduling in a backscatter communication (BAC)-non-orthogonal multiple access (NOMA)<sup>98</sup> internet-of-things (IoT) network. They jointly optimize the transmit power of downlink IoT<sup>99</sup> users and the reflection coefficient of uplink backscatter devices using SAC. The proposed algorithm guarantees the quality of service requirements of downlink users while simultaneously enhancing the sum rate of uplink backscatter devices. Numerical outcomes indicate that the proposed algorithm performs better than conventional optimization approaches in terms of the average sum rate of uplink backscatter devices. In Reference [100],

the authors explore cooperative RA in cellular networks where wireless information and power transfer are performed concurrently in a time-varying channel environment. They utilize SAC to determine a feasible RA strategy that maximizes data rate and system fairness while minimizing the channel switching penalty. To address the limited empirical dataset of SAC, they introduce two augmentation schemes for the experience replay buffer of SAC. Simulation findings reveal that the proposed schemes benefit the training model and effectively improve the performance of the algorithm. Moreover, in Reference [101], the researchers propose an innovative approach, the Constrained Discrete-Continuous Soft Actor-Critic (CDC-SAC) algorithm, to address a RA problem with a mixed action space. This action space encompasses both discrete channel allocation and continuous energy harvesting time division while adhering to constraints related to energy consumption and queue package length. To effectively address the challenge of RA using DRL, they propose innovative modifications to both the network architecture and the process of policy learning. The simulations conducted demonstrate remarkable performance enhancements achieved by the proposed algorithm, particularly in terms of total throughput, all while satisfying stringent constraints. Table 3 summarizes each algorithm's description and application,<sup>102,103</sup> making it easier for readers to understand the different DRL algorithms used in RA in various 5G C-RAN scenarios.

When implementing a DRL, specifically in the context of C-RAN, there are several aspects to consider. Here is an overview of the implementation aspects:

- *Define the problem:* Clearly define the problem you want to solve using reinforcement learning. In the case of C-RAN, this might involve optimizing the resource allocation, such as the allocation of computational resources and bandwidth, to maximize the overall performance of the network.
- *Design the state, action, and reward space:* Define the state space, which represents the current state of the network, including information such as user traffic, network congestion, and available resources. Define the action space, which represents the possible actions the agent can take, such as allocating resources to different network elements. Lastly, define the reward function, which quantifies the performance of the agent based on the actions taken in a certain state.
- *Choose a reinforcement learning algorithm:* Select a suitable reinforcement learning algorithm to train your agent. Popular algorithms include Q-learning,<sup>104</sup> Deep Q-Networks (DQN), and Proximal Policy Optimization (PPO). Consider the specific requirements and characteristics of the C-RAN problem when choosing the algorithm.
- *Implement the environment:* Develop an environment that simulates the C-RAN setup. This environment should provide the necessary functionality to interact with the agent, including methods for transitioning between states, taking actions, and calculating rewards. Ensure that the environment accurately reflects the dynamics of the C-RAN system.
- *Train the agent:* Use the chosen reinforcement learning algorithm to train the agent. This involves iteratively updating the agent's policy based on interactions with the environment. Implement the training loop, which involves selecting actions, updating the agent, and collecting experience.
- *Fine-tune and optimize:* Experiment with different hyperparameters and techniques to improve the agent's performance. This may involve adjusting exploration-exploitation trade-offs, adjusting the learning rate, or applying techniques like experience replay or reward shaping.
- *Evaluation and deployment:* Once the agent has been trained, evaluate its performance using appropriate metrics in realistic or simulated scenarios. If the performance is satisfactory, deploy the trained agent in the C-RAN system and monitor its performance in real-world conditions.
- *Consider scalability and computational efficiency:* C-RAN systems often deal with large-scale networks and require real-time decision-making. Therefore, it is important to consider the scalability and computational efficiency of your reinforcement learning algorithm. This may involve optimizing the training process, using parallel, or adopting algorithms specifically designed for large-scale problems.
- *Handle network dynamics:* C-RAN systems are subject to constant changes and demands. It is crucial to design your reinforcement learning algorithm to handle network dynamics effectively. This may involve incorporating time-series data, adapting the state and action spaces dynamically, or using techniques like online learning to adapt the agent's policy in real time.
- *Integration with existing network infrastructure:* When implementing a reinforcement learning algorithm in C-RAN, it is important to consider the integration of existing network infrastructure. This may involve developing interfaces to communicate with network components, such as base stations and central processing units, and ensuring compatibility with existing management and control systems.

**TABLE 3** Cutting-edge DRLs are utilized for RA.

| Algorithm       | Description                                                                                                                                                                                                                                                                             | Application                                                                                                                                                                                                                                                                                                                                      | Computational complexity |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| DQN             | RL algorithm that uses NN to represent the Q function; estimates the long-term reward for taking a particular action in a given state                                                                                                                                                   | Heterogeneous Cloud Radio Access Networks (H-CRAN), intelligent reflecting surface (IRS) enhanced orthogonal frequency division multiplexing (OFDM) systems, RAN containing IIoT devices, C-RAN                                                                                                                                                  | $O(n * m * t)$           |
| Policy Gradient | Policy optimization, directly optimizing the mapping from states to actions, proves more efficient for continuous action space problems.                                                                                                                                                | Content distribution in fog radio access networks, eMBB and URLLC coexistence in 5G                                                                                                                                                                                                                                                              | $O(N * T * d)$           |
| Actor-Critic    | Incorporating elements from both value-based and policy-based approaches, the “actor” component learns to make action choices according to the existing policy, while the “critic” component assesses the policy’s effectiveness.                                                       | Multi-user MEC system, HetNets powered by hybrid energy, fog-enabled IoT                                                                                                                                                                                                                                                                         | $O(k * n)$               |
| A3C             | A distributed variant of actor-critic algorithms; multiple agents learn independently and asynchronously in parallel; use the advantage function to measure how much better or worse an action is compared to the average action                                                        | RA in 5G for RAN slicing, video streaming over software-defined mobile networks combined with MEC                                                                                                                                                                                                                                                | $O(n)$                   |
| PPO             | Optimizes the objective function subject to a constraint on the amount by which the policy can change; ensures that the policy update does not deviate too far from the previous policy                                                                                                 | Base stations and mobile devices, automated warehouse scenario using swarm robotics controlled by an industrial controller over the 5G network, multi-UAV collaborative QoS management during epidemics or disasters                                                                                                                             | $O(n)$                   |
| SAC             | An actor-critic method that seeks to maximize the expected reward while maintaining a minimum level of entropy in the policy. The term “soft” refers to the fact that SAC optimizes a lower bound on the standard RL objective, which includes an entropy term to encourage exploration | MEC system for cooperative computation offloading among edge-cloud and edge-edge servers, dynamic resource scheduling in a BAC-NOMA, cooperative RA in cellular networks with simultaneous wireless information and power transfer in a time-varying channel environment                                                                         | $O(n)$                   |
| DDPG            | An actor-critic method for continuous control problems. The “actor” outputs a deterministic action given the current state, while the “critic” evaluates the quality of that action.                                                                                                    | Optimal utilization of resources and effective computation offloading in Fog Access Points (F-APs) Task offloading and efficient RA in Fog Radio Access Networks (F-RANs) Enhancing downlink sum rate while considering cost factors through integrated optimization of Radio Access Technologies (RATs) assignment and PA in advanced HetNets.  | $O(n^2)$                 |
| TD3             | Builds upon the success of DDPG with improvements in reducing overestimation and variance in Q-value estimates. Uses two critics instead of one and applies random noise to the target policy during training to encourage exploration                                                  | PA in the IoRT network—Improving access success probability of 5G-and-beyond low earth orbit (LEO) satellite communication systems Jointly optimizing task segmentation and PA to minimize total energy consumption in MEC systems Equilibrated and fast RA algorithm for massive and diversified MTC services using a multi-agent DRL approach. | $O(n)$                   |

- *Continuous learning and adaptation:* C-RAN systems require continuous learning and adaptation to stay optimized under changing conditions. Consider implementing mechanisms to periodically retrain the agent using updated data or incremental learning techniques to adapt the policy over time.
- *Robustness and safety considerations:* Reinforcement learning algorithms should be designed with robustness and safety in mind. Consider techniques such as reward shaping, constraint handling, or safety mechanisms to ensure that the agent operates within desired bounds and avoids harm.
- *Monitoring and Implementing mechanisms* to monitor and validate the performance of the reinforcement learning agent in the C-RAN system. This may include logging key metrics, conducting statistical analysis, or comparing the agent's performance against baseline or expert systems.
- *Iterative improvement:* Even after deployment, continue to iterate and improve the reinforcement learning algorithm based on feedback and real-world performance. Monitor the system's behavior, analyze any shortcomings or unexpected outcomes, and refine the algorithm accordingly. Implementing a reinforcement learning algorithm in C-RAN requires a multidisciplinary approach, combining knowledge in networking, machine learning, and system design. It is essential to have a clear understanding of the problem, design appropriate algorithms and architectures, and carefully validate and monitor the performance of the system.

### 3 | CHALLENGES AND SOLUTIONS

#### 3.1 | Scalability issues in applying DRL to C-RAN RA

DRL has gained significant attention as a highly promising technique for effectively addressing Radio Access (RA) challenges in the context of C-RANs. However, scaling DRL-based RA to large-scale networks poses significant challenges. In this context, there are several scalability issues related to the measurement of DRL's efficacy in C-RAN RA. The availability of abundant training data is crucial for effectively training DRL models.<sup>105</sup> However, acquiring such data in large-scale networks can pose substantial challenges due to the extensive computational resources required. Furthermore, the feasibility of implementing a solution depends on both the complexity of the specific problem being addressed and the scale of the network under consideration.<sup>106</sup> As the scale of the network increases, the complexity of the problem also increases, posing further challenges to DRL's scalability. Another critical issue is the need to design a suitable reward function that can accurately capture the objectives of the C-RAN RA problem.<sup>107</sup> Designing such a reward function is crucial to ensure that the DRL model learns effective RA strategies. However, defining a good reward function can be challenging, particularly when dealing with complex problems and multiple conflicting objectives.<sup>108</sup> To address these scalability issues, researchers have proposed various techniques, including transfer learning,<sup>109,110</sup> regularization,<sup>111</sup> and hierarchical learning.<sup>112</sup> Transfer learning involves using pre-trained models to bootstrap the learning process, which can significantly reduce the amount of required training data.<sup>113</sup> Regularization techniques can help prevent overfitting and improve generalization performance, while hierarchical learning can enable the decomposition of complex problems into smaller, more manageable sub-problems.

In conclusion, DRL can offer significant benefits for RA in C-RAN, but its scalability is a significant challenge. Addressing scalability issues requires a combination of advanced techniques, including transfer learning, regularization, and hierarchical learning. By employing these techniques, it may be possible to develop efficient and scalable DRL-based RA algorithms for C-RANs.

#### 3.2 | Convergence challenges of DRL algorithms in C-RAN

Another of the main challenges is the large number of parameters that need to be tuned in the DRL models.<sup>114,115</sup> This can make it difficult to converge to an optimal solution, especially when dealing with complex networks with a large number of nodes.<sup>116</sup> In addition, the high-dimensional state representation required by many DRL models can also pose a convergence challenge.<sup>117</sup>

Another challenge is the non-stationarity of the environment in C-RAN. The dynamics of the network change over time due to various factors such as user mobility, traffic variability, and channel fading. This makes it difficult to develop a DRL policy that can adapt to these changes and converge to an optimal solution.<sup>118</sup>

To address these challenges, researchers are exploring different approaches such as using ensemble learning to improve the stability and robustness of DRL algorithms,<sup>119</sup> incorporating domain knowledge into the DRL models, and developing new techniques for exploration and exploitation in the non-stationary environment of C-RAN. Overall, while there are certainly challenges associated with using DRL algorithms in C-RAN, there is also significant promise for improving network performance and efficiency through these methods.

### 3.3 | Ensuring fairness in RA with DRL algorithms

DRL uses trial-and-error-based learning to optimize decision-making processes. While DRL algorithms have shown promising results in various applications such as resource management<sup>120</sup> and network slicing,<sup>121</sup> there are some challenges in ensuring fairness and justice in the allocation of resources using these algorithms.

One of the main challenges is the potential for bias in the data used to train the algorithm, which can lead to unfair outcomes.<sup>122</sup> For example, if the data used to train the algorithm is biased towards a certain group or demographic, the algorithm may learn to favor that group over others, leading to unequal allocation of resources.

Another challenge is the lack of transparency in how the algorithm makes decisions. DRL algorithms can be very complex, with many layers of NNs making decisions based on input data.<sup>123</sup> It can be difficult to understand how the algorithm arrived at a particular decision, making it challenging to identify and correct any biases or unfairness.

To tackle these challenges, researchers have put forth diverse methodologies and approaches such as incorporating fairness constraints into the objective function of the algorithm,<sup>124</sup> carefully selecting and preprocessing the training data to reduce bias, and using explainable AI techniques to better understand how the algorithm makes decisions.<sup>125</sup>

### 3.4 | Security and privacy

One of the main challenges for RA considering security<sup>104</sup> and privacy in C-RAN with DRL algorithms arises due to the sensitive nature of the data involved in the network. This includes user information, network parameters, and communication patterns. DRL algorithms, while capable of optimizing resource allocation, often require access to a significant amount of data to learn from. This can lead to privacy concerns as exposing such data to the algorithm may risk unauthorized access or misuse.<sup>126</sup> To address this challenge in the future, several approaches can be considered:

- *Privacy-preserving algorithms*: Research can focus on developing DRL algorithms that are privacy-preserving by design. This involves techniques such as secure multi-party computation, differential privacy, and federated learning. These approaches ensure that sensitive data remains encrypted or aggregated, thereby protecting privacy.<sup>127</sup>
- *Data anonymization*: Implementing techniques to anonymize the data used for training the DRL algorithms can be beneficial. This involves removing or obfuscating personally identifiable information, while still providing enough information for the algorithm to learn and make efficient RA decisions.<sup>128</sup>
- *Secure communication protocols*: Designing secure communication protocols within the C-RAN infrastructure can help protect against unauthorized access and data breaches. Techniques such as encryption, authentication, and access control mechanisms can be employed to ensure the privacy and security of the data transmitted between different components of the network.
- *Regulatory frameworks and policies*: Establishing robust regulatory frameworks and policies that govern the collection, storage, and usage of data in C-RANs can provide a strong foundation for security and privacy. These frameworks should address issues related to data ownership, and consent, and provide guidelines for ensuring data protection while leveraging DRL algorithms for resource allocation.

Overall, addressing the challenge of RA considering security and privacy in C-RANs with deep reinforcement learning algorithms requires a multi-faceted approach involving algorithmic advancements, data privacy techniques, secure communication protocols, and regulatory measures.

## 4 | FUTURE RESEARCH DIRECTIONS

### 4.1 | Open areas for future investigation in the application of DRL to C-RAN RA

There are several areas for future investigation in the application of DRL to C-RAN RA. Some potential research directions include:

- *Network dynamics modeling*: Developing more accurate and realistic models to capture the complex dynamics of C-RAN networks. This includes modeling the interactions among different network elements, such as base stations, central processing units, and fronthaul links.
- *Reward design*: Designing appropriate reward functions that can effectively capture the performance objectives of C-RAN RA. This involves considering different performance metrics, such as energy efficiency, spectral efficiency, and quality of service, and finding the right trade-offs between them.
- *Exploration-exploitation trade-off*: Investigating different exploration-exploitation strategies in DRL algorithms for C-RAN resource allocation. This includes exploring new techniques to balance the exploration of unexplored actions with the exploitation of known good actions.
- *Scalability*: Addressing the scalability challenges associated with applying DRL to large-scale Cloud-RAN networks. This involves developing techniques that can handle a large number of base stations, users, and resource blocks while maintaining good computational efficiency.
- *Transfer learning*: Exploring the use of transfer learning techniques to leverage pre-trained DRL models in Cloud-RAN RA. This includes investigating how to transfer knowledge from one network scenario to another, or from one application domain to another, to accelerate the learning process.
- *Robustness and fairness*: Addressing the robustness and fairness issues in DRL-based C-RAN RA. This includes investigating how to design algorithms that can adapt to changes in network conditions and ensure fair RA among different users.
- *Real-time implementation*: Investigating the feasibility of implementing DRL-based RA algorithms in real-world C-RAN systems. This includes considering the computational requirements, communication overhead, and the availability of training data in a real-time setting.

Overall, these areas of investigation can help advance the application of DRL to C-RAN RA and contribute to more efficient and reliable future wireless networks.

### 4.2 | Potential impact of DRL on the development of 5G networks

Transfer learning is a promising area for applying DRL to C-RAN RA in 5G networks. This methodology entails utilizing acquired knowledge from one particular task to enhance the performance of a related task. By training an agent on data from one network and transferring that knowledge to another network, it may be possible to reduce the volume of data necessary for agent training and performance enhancement.

Multi-agent reinforcement learning (MARL) is another area worth exploring for C-RAN RA in 5G networks. In MARL, multiple agents work together to achieve a shared objective. For instance, in C-RAN RA, agents could collaborate to optimize overall network efficiency instead of solely focusing on their resources.

DRL algorithms have demonstrated success in resolving problems in static environments but face challenges in dynamic environments where there are frequent changes. Researching how to apply DRL to dynamic network environments is a vital area for improving C-RAN RA in 5G networks.

As 5G mobile networks are projected to support devices with varying requirements and capabilities, investigating how to apply DRL to RA in heterogeneous networks could ensure the effective allocation of network resources across all devices.

The potential impact of DRL on 5G networks is significant. Firstly, using DRL to optimize RA in C-RANs can potentially enhance overall network efficiency, which results in better performance and reduced costs. Secondly, DRL algorithms can learn to allocate resources effectively, leading to increased network capacity and improved support for

high-bandwidth applications. Thirdly, DRL algorithms can optimize RA to reduce energy consumption, particularly crucial for mobile networks where battery life is a concern. Finally, the application of DRL algorithms enables the creation of autonomous networks capable of dynamically adjusting to varying circumstances and maximizing their operational efficiency without the need for human intervention.

## 5 | CONCLUSION

In recent years, the emergence of 5G wireless communication networks has created a need for highly efficient resource allocation mechanisms to support the increasing demands of diverse and complex applications. A cloud radio access network (C-RAN) has been identified as a promising solution for meeting these demands by centralizing radio access network functions in a cloud-based processing unit. However, traditional resource allocation schemes may not be scalable enough to handle the dynamic nature of 5G networks. In this context, deep reinforcement learning (DRL) algorithms offer an alternative approach to resource allocation that uses machine learning techniques to learn complex decision-making policies directly from raw data without explicit human intervention.

This paper reviews recent research on resource allocation in 5G C-RAN using DRL algorithms. The review provides insights into the potential solutions and challenges in applying DRL to C-RAN resource allocation, including scalability issues, convergence challenges, and ensuring fairness in resource allocation. The paper also discusses state-of-the-art DRL algorithms applied to C-RAN resource allocation and identifies open areas for future investigation in the application of DRL to C-RAN resource allocation. Overall, this review highlights the potential of DRL algorithms in optimizing resource allocation in 5G C-RAN and their potential impact on the development of 5G networks.

## CONFLICT OF INTEREST STATEMENT

The authors of this research paper declare no conflict of interest.

## DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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**How to cite this article:** Khani M, Jamali S, Sohrabi MK, Sadr MM, Ghaffari A. Resource allocation in 5G cloud-RAN using deep reinforcement learning algorithms: A review. *Trans Emerging Tel Tech*. 2024;35(1):e4929. doi: 10.1002/ett.4929